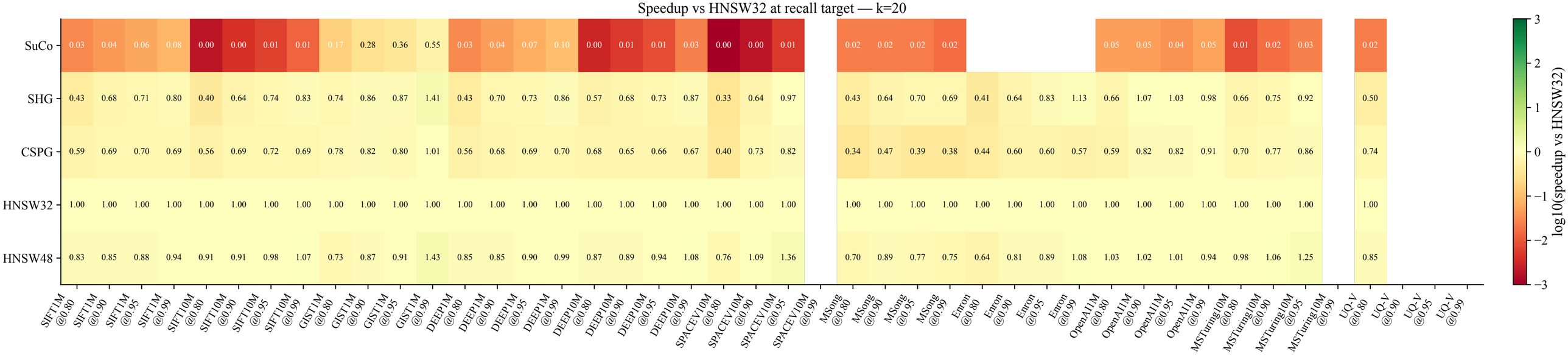

Speedup vs HNSW32 at recall target — k=20
SuCo
SHG
CSPG
HNSW32
HNSW48
0.03 0.04 0.06 0.08 0.00 0.00 0.01 0.01 0.17 0.28 0.36 0.55 0.03 0.04 0.07 0.10 0.00 0.01 0.01 0.03 0.00 0.00 0.01 0.02 0.02 0.02 0.02 0.05 0.05 0.04 0.05 0.01 0.02 0.03 0.02
0.43 0.68 0.71 0.80 0.40 0.64 0.74 0.83 0.74 0.86 0.87 1.41 0.43 0.70 0.73 0.86 0.57 0.68 0.73 0.87 0.33 0.64 0.97 0.43 0.64 0.70 0.69 0.41 0.64 0.83 1.13 0.66 1.07 1.03 0.98 0.66 0.75 0.92 0.50
0.59 0.69 0.70 0.69 0.56 0.69 0.72 0.69 0.78 0.82 0.80 1.01 0.56 0.68 0.69 0.70 0.68 0.65 0.66 0.67 0.40 0.73 0.82 0.34 0.47 0.39 0.38 0.44 0.60 0.60 0.57 0.59 0.82 0.82 0.91 0.70 0.77 0.86 0.74
1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
0.83 0.85 0.88 0.94 0.91 0.91 0.98 1.07 0.73 0.87 0.91 1.43 0.85 0.85 0.90 0.99 0.87 0.89 0.94 1.08 0.76 1.09 1.36 0.70 0.89 0.77 0.75 0.64 0.81 0.89 1.08 1.03 1.02 1.01 0.94 0.98 1.06 1.25 0.85
SIFT1M @0.80
SIFT1M @0.90
SIFT1M @0.95
SIFT1M @0.99
SIFT10M @0.80
SIFT10M @0.90
SIFT10M @0.95
SIFT10M @0.99
GIST1M @0.80
GIST1M @0.90
GIST1M @0.95
GIST1M @0.99
DEEP1M @0.80
DEEP1M @0.90
DEEP1M @0.95
DEEP1M @0.99
DEEP10M @0.80
DEEP10M @0.90
DEEP10M @0.95
DEEP10M @0.99
SPACEV10M @0.80
SPACEV10M @0.90
SPACEV10M @0.95
SPACEV10M @0.99
MSong @0.80
MSong @0.90
MSong @0.95
MSong @0.99
Enron @0.80
Enron @0.90
Enron @0.95
Enron @0.99
OpenAI1M @0.80
OpenAI1M @0.90
OpenAI1M @0.95
OpenAI1M @0.99
MSTuring10M @0.80
MSTuring10M @0.90
MSTuring10M @0.95
MSTuring10M @0.99
UQ-V @0.80
UQ-V @0.90
UQ-V @0.95
UQ-V @0.99
log10(speedup vs HNSW32)
3
2
1
0
−1
−2
−3